

Appendix A

The vernier caliper

The vernier caliper is an instrument that can make very accurate linear measurements. It was introduced in 1631 by Pierre Vernier of France. It utilises two graduated scales: a main scale similar to that found on a ruler and another graduated auxiliary scale, the vernier scale, that slides parallel to the main scale and enables readings to be made to a fraction of a division on the main scale. The webpage

<http://www.phy.uct.ac.za/courses/phylab1/vernier>

gives more information about the use of the vernier caliper.

The vernier caliper has two sets of jaws, one for measuring outer measurements (like the outer diameter of a disc) and one for inner measurements (like the inside diameter of a pipe). It also has a pin for measuring depths (like the depth of a groove) - see **figure 1**.

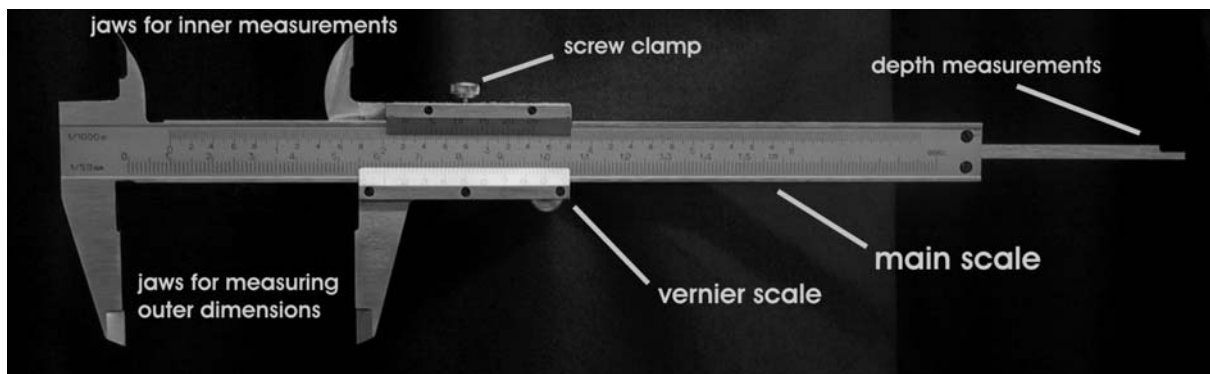


Fig. 1 a Vernier Caliper.

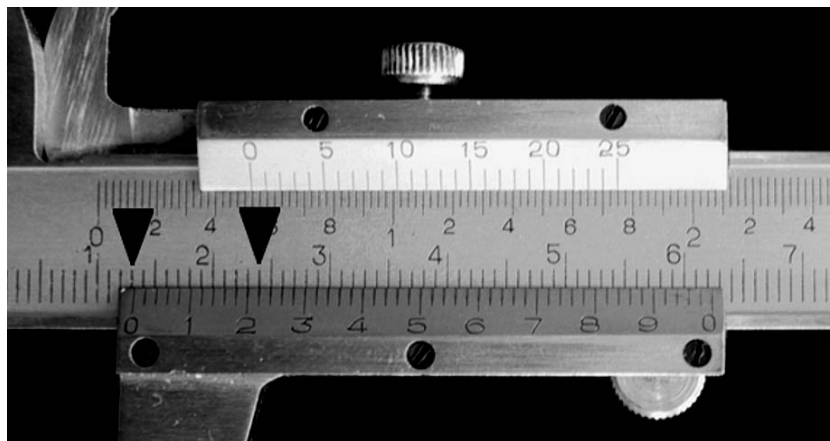


Fig. 2 Taking a reading on a vernier caliper.

The caliper in **figure 2** has 50 divisions on the vernier scale. The leftmost one, the index line, indicates the reading on the main scale, 13 mm in this case since the index line is just to the right of it. The reading on the vernier scale is taken at the position where a vernier line lines up with a line on the main scale. This is at the line just to the right of the 2 line on the vernier scale. This corresponds to a vernier reading of 0,22 mm. The caliper reading is therefore 13,22 mm (for all three types of measurements - outer, inner and depth measurements).

The meaning of the vernier divisions can be inferred by closing the caliper completely and noting that the 50 divisions on it are 49 mm long. One vernier division is therefore 0,98 mm wide. The first vernier division is therefore 0,02 mm short of the first millimeter mark on the main scale, while the second one is 0,04 mm short of the second millimeter mark on the main scale. If the jaws were therefore to be opened 0,02 mm the first vernier line will coincide with first millimeter line on the main scale, and if it were to be opened 0,04 mm the second vernier line will coincide with the second line on the main scale...

In **figure 3** the outer diameter of a cylinder is measured. The jaws were closed on it and the clamping screw tightened.

The smallest division on the vernier in **figure 3** is 0,02mm shorter than one millimeter.

The reading on the main scale ① is 7mm

The reading on the vernier scale ② is 0,28mm

The diameter of the cylinder is therefore 7,28mm

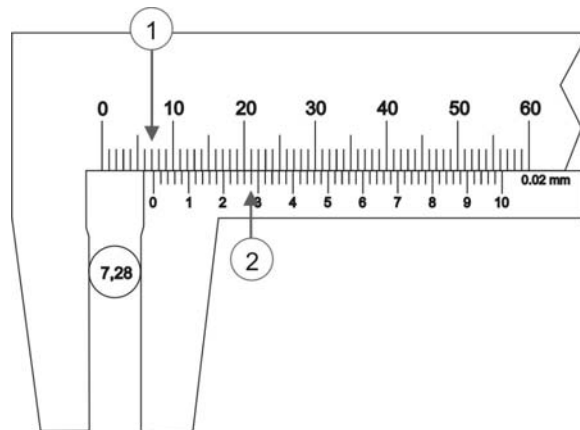


Fig. 3 Measuring with a vernier caliper

In order to determine the diameter it is therefore necessary to take readings on both the main- and vernier scales and combine these two readings. The vernier divisions of the caliper in **figure 4** are 0,05 mm shorter than two

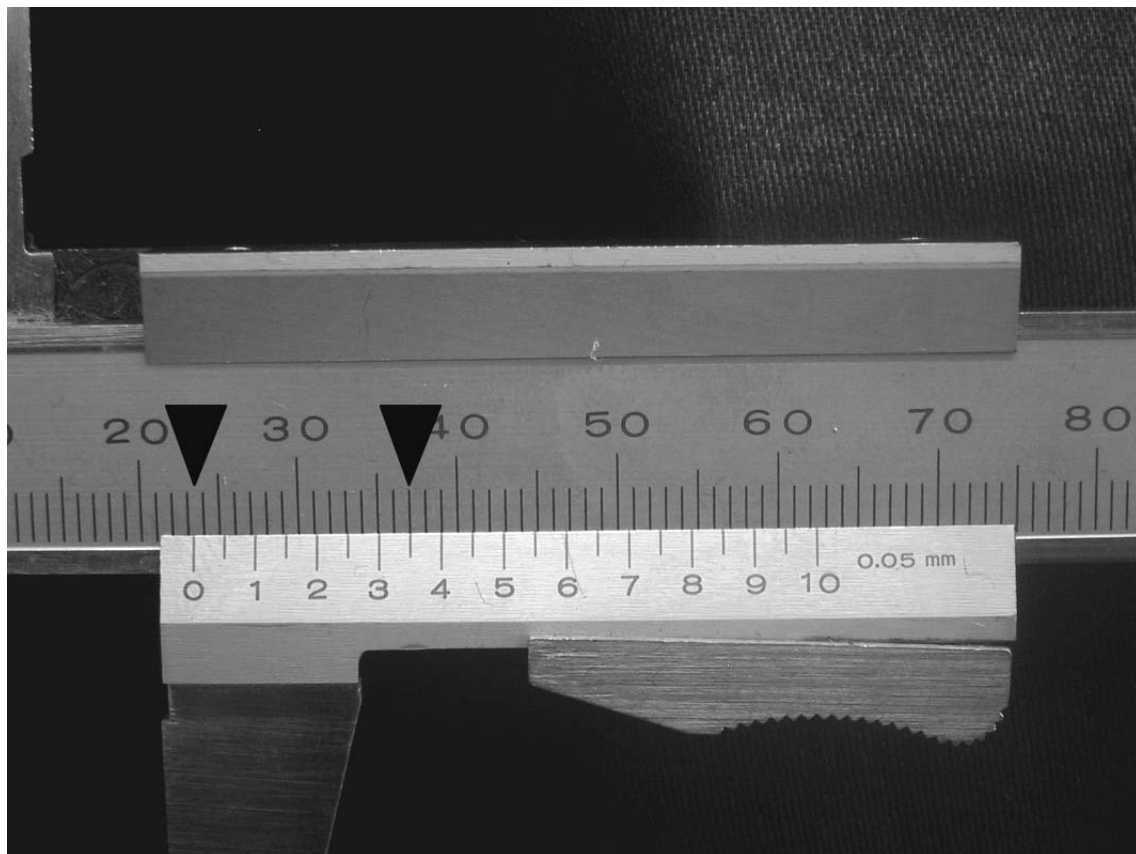


Fig. 4 Taking a reading on a vernier caliper.

millimeters. The reading on the main scale is 23 mm, while the vernier reading is 0,35 mm. The caliper reading is therefore 23,35 mm.

In **figure 5** another measurement is made with a similar caliper:

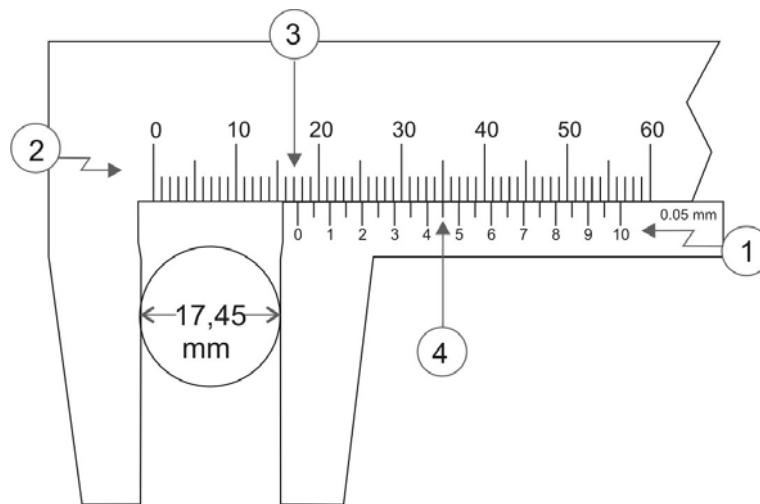


Fig. 5 Reading of a vernier caliper

The vernier ① is the auxiliary scale that slides along the main scale ②.

The reading on the main scale is taken just to the left of the zero on the vernier scale ③ (17mm in this case).

The reading on the vernier is taken where the marks on the vernier scale and the main scale coincide ④ (0,45mm in this case).

The final reading is the sum of readings on the vernier and the main scale.

The diameter of the object is therefore 17,45mm.
